

Breast Cancer Tumor Classification

AUG| 2023

Ultra Sound Team



Maen Qawasmeh

Mohammad Alkarmy

Roaa Ma’moun

Nisreen Alomari

Table Of Content



01. Introduction

02. EDA

03. Data Preprocessing

04. Feature Extraction

05. Models’ Experiments

06. Learning Curves

07. Hyperparameter Tuning

08. Future Work

09. Conclusion

Introduction



Introduction

Breast cancer is cancer that forms in the breast tissue. Among all cancer deaths worldwide, many women are dying, majorly due to breast cancer.
This challenge aims to recognize and classify breast photos and determine the type of tumor present, whether benign or malignant, if present.

EDA

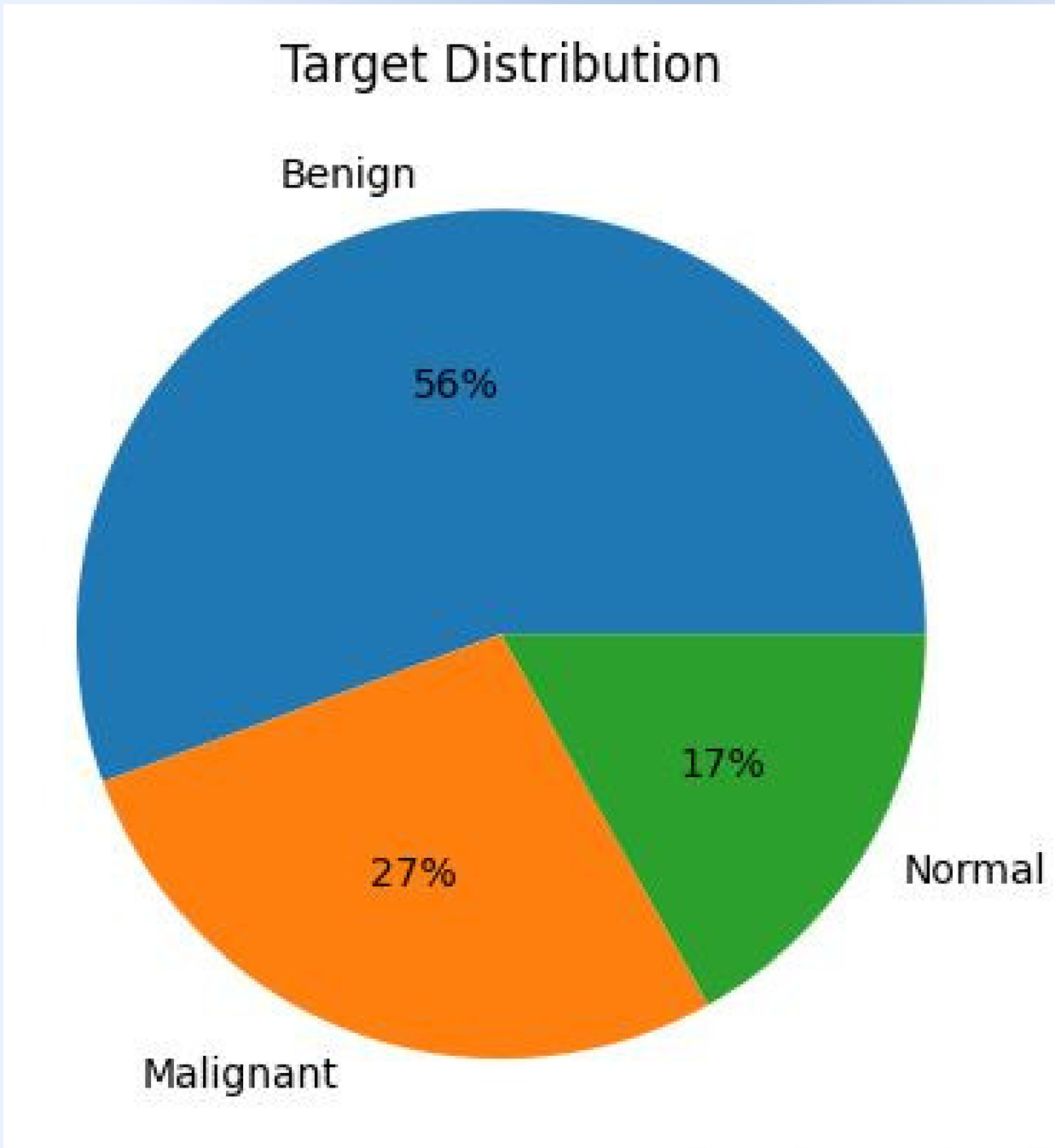
EDA

Target Distribution:

91 samples for the Normal type of breast cancer

148 samples for the Malignant type of breast cancer

300 samples for the Benign type of breast cancer



Data Preprocessing

Resizing & Cropping
Data Augmentation
Image Filtering



1

Resizing & Cropping

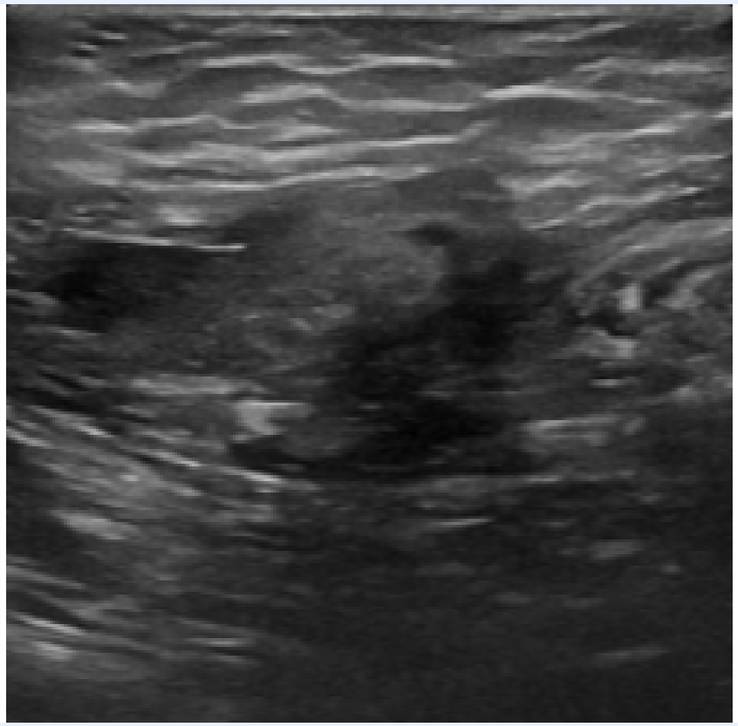
RESIZING: CHANGES THE DIMENSIONS OF THE IMAGE, WHICH USUALLY AFFECTS THE FILE SIZE.

CROPPING: INVOLVES CUTTING AWAY PART OF THE ORIGINAL IMAGE AND RESULTS IN SOME OF THE PIXELS BEING DISCARDED.



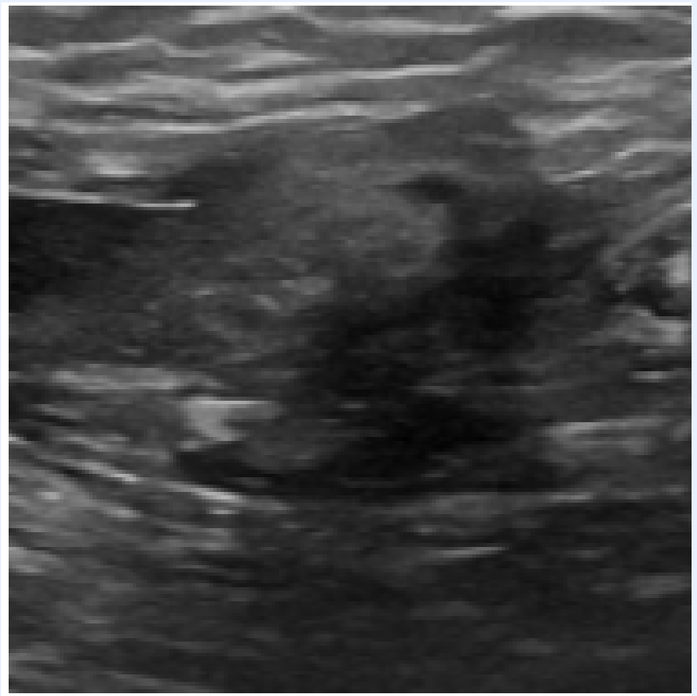
Image Classified as Normal Tumor

Resizing



200 X 200

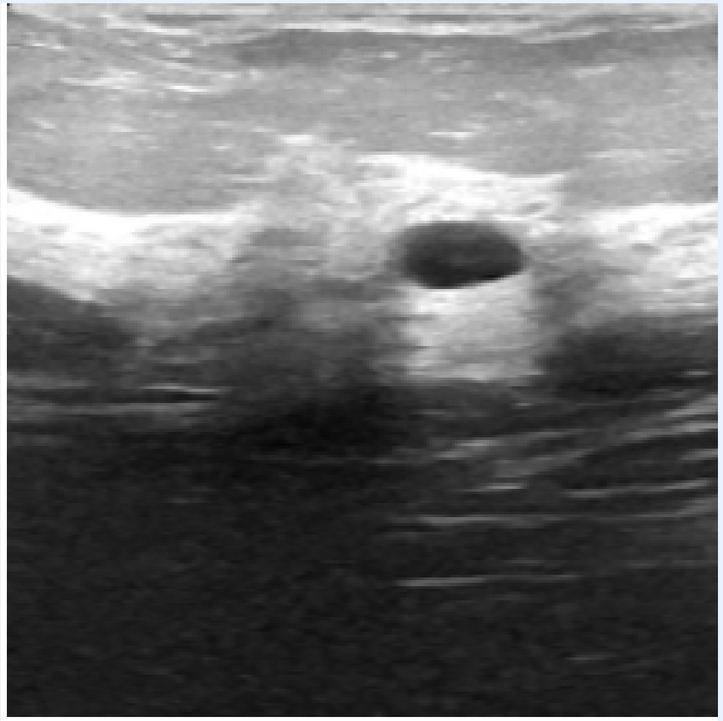
Cropping



150 X 150

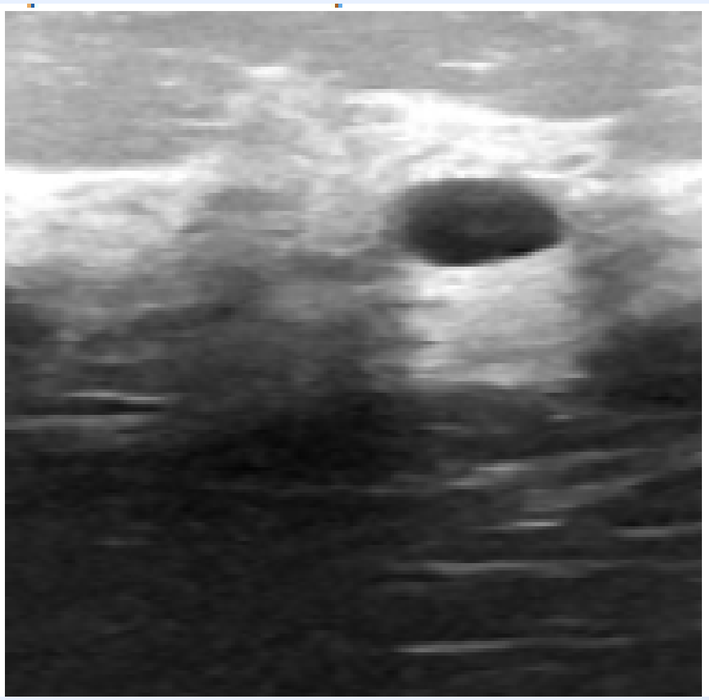
Image Classified as Benign Tumor

Resizing



200 X 200

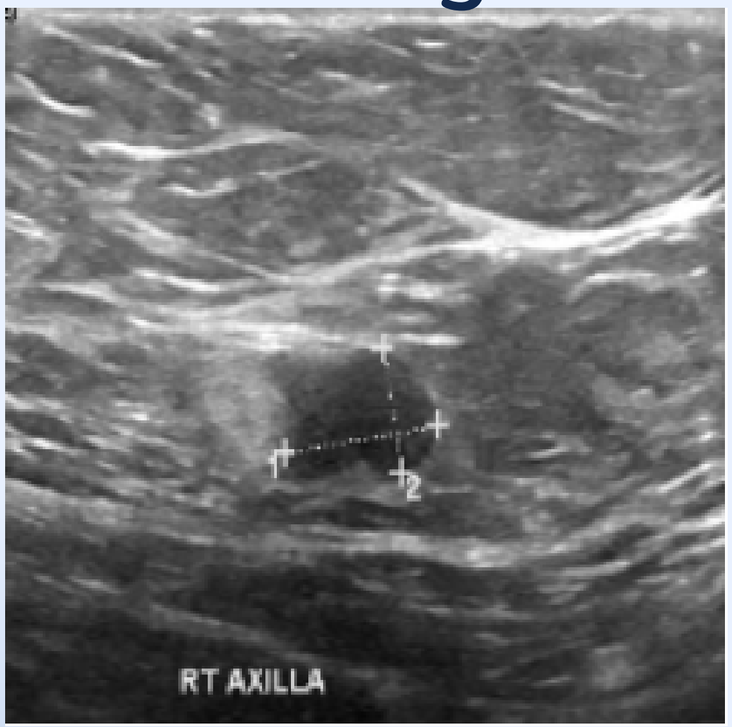
Cropping



150 X 150

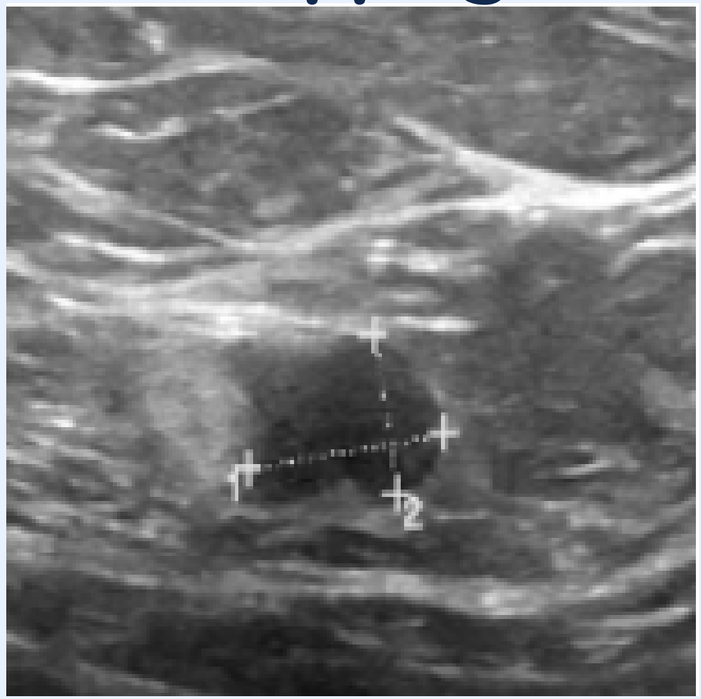
Image Classified as Malignant Tumor

Resizing



200 X 200

Cropping



150 X 150

2

Data Augmentation

DATA AUGMENTATION: IS A PROCESS OF ARTIFICIALLY INCREASING THE AMOUNT OF DATA BY GENERATING NEW DATA POINTS FROM EXISTING DATA. THIS INCLUDES ADDING MINOR ALTERATIONS TO DATA OR USING MACHINE LEARNING MODELS TO GENERATE NEW DATA POINTS IN THE LATENT SPACE OF ORIGINAL DATA TO AMPLIFY THE DATASET.



Data Augmentation

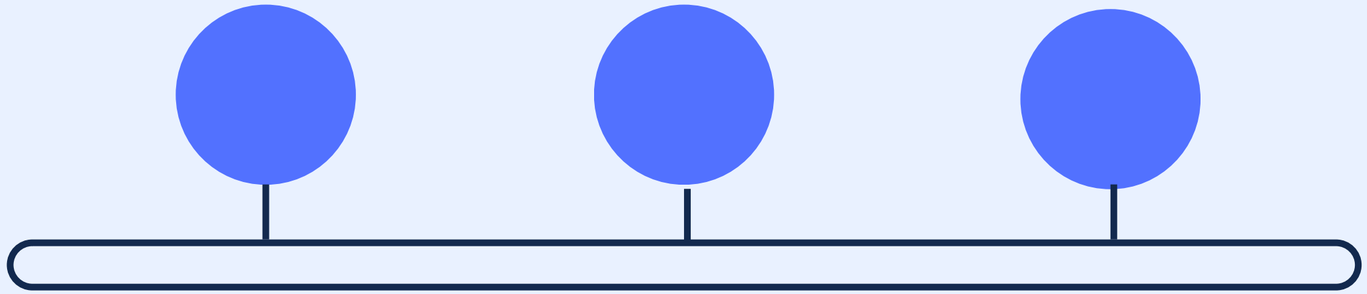
- The Data Augmentation process starts by categorizing each image into its appropriate group of 'Malignant,' 'Benign,' or 'Normal.'
- This initial step allows us to perform focused data augmentation on specific classes, achieving a balanced distribution across all three. To enhance our classification model's overall effectiveness.

Data Augmentation

Data Augmentation Method / **Needed Samples**	Reflection Using cv2.wrapperspective	Augmenter
209 to be classified as a Normal Tumor	x-axis Reflection y-axis Reflection x-axis & y-axis Reflection	Multiple Augmentation Methods (**Zooming & Rotation**) were applied to images to produce 209 other samples.
152 to be classified as a Malignant Tumor	x-axis Reflection y-axis Reflection	Multiple Augmentation Methods (**Shining**) were applied to images to produce 152 other samples.

3

Image Filtering Techniques



1st

Bilateral filter image

2nd

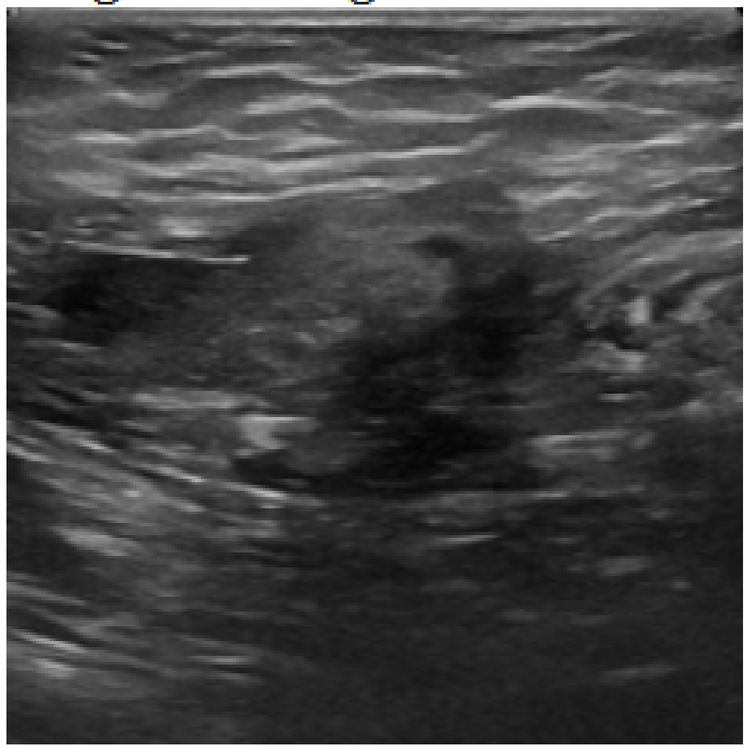
Schar filter image

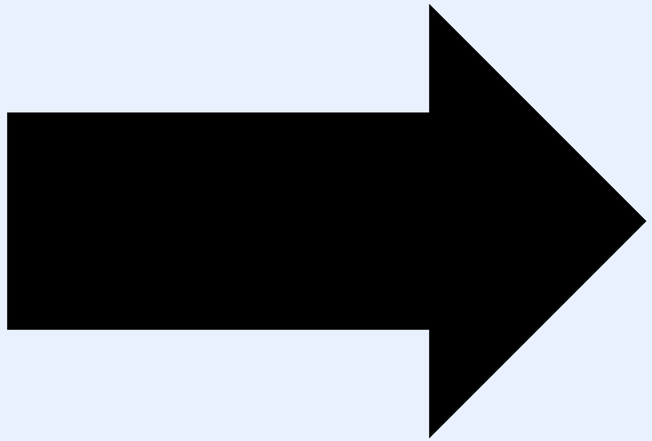
3rd

Gabor filter image

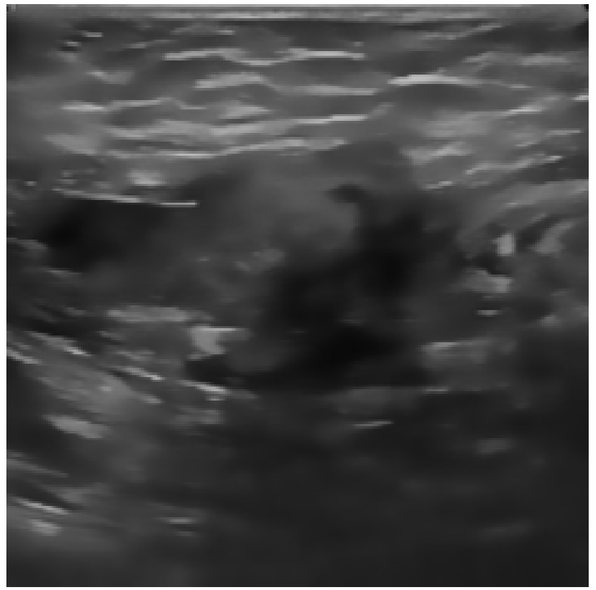
Normal Tumor

Original Image

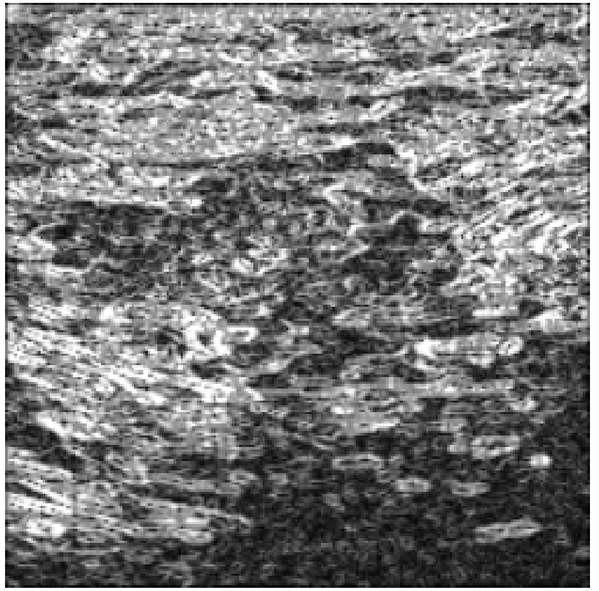




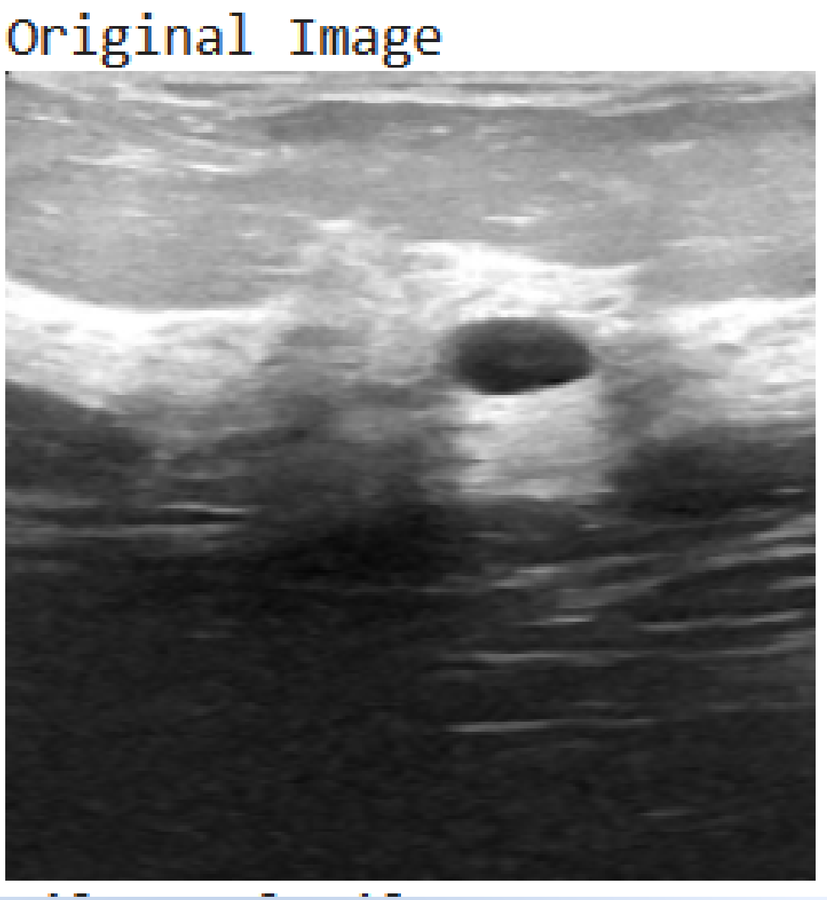
Bilateral Filter Image

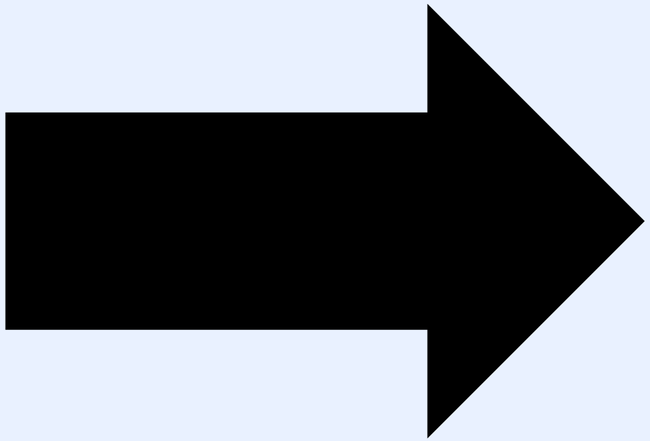


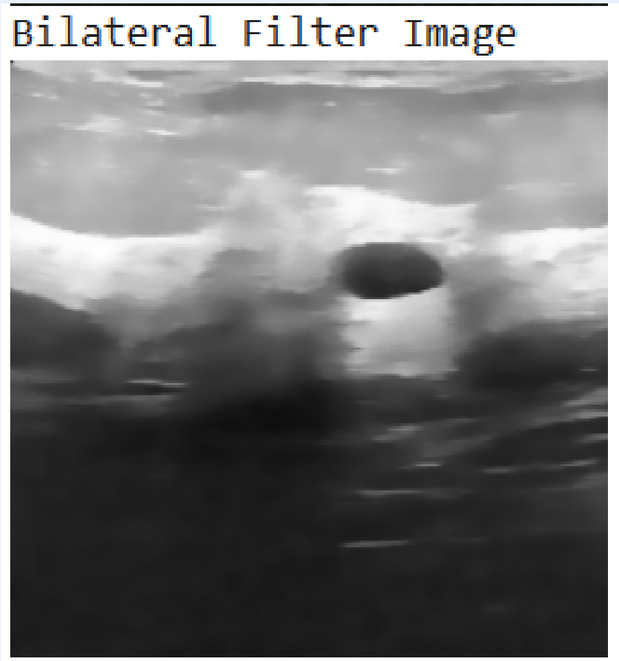
Scharr Filter Image

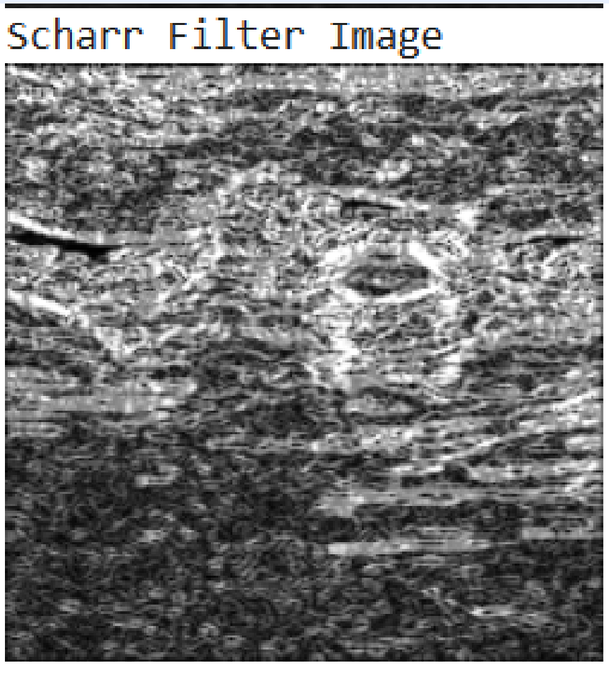


Benign Tumor







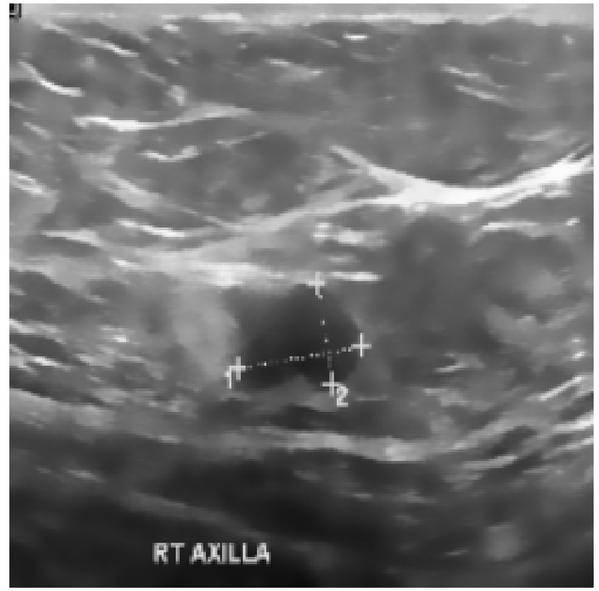


Malignant Tumor

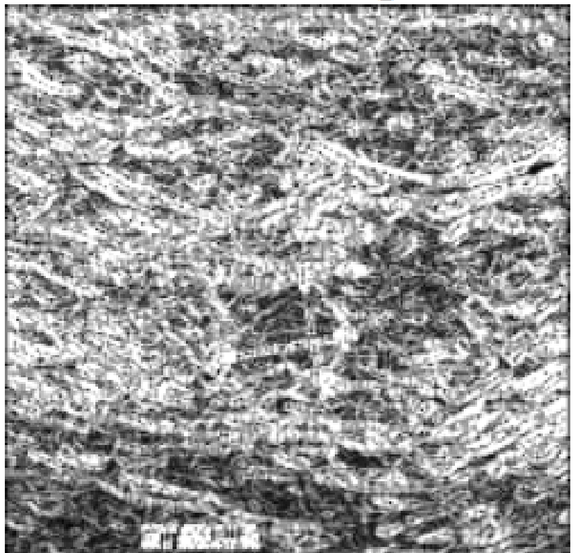
Original Image



Bilateral Filter Image

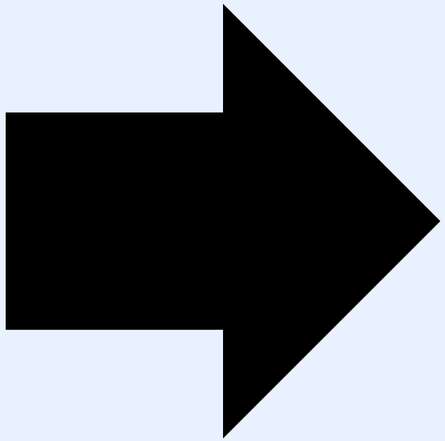


Scharr Filter Image



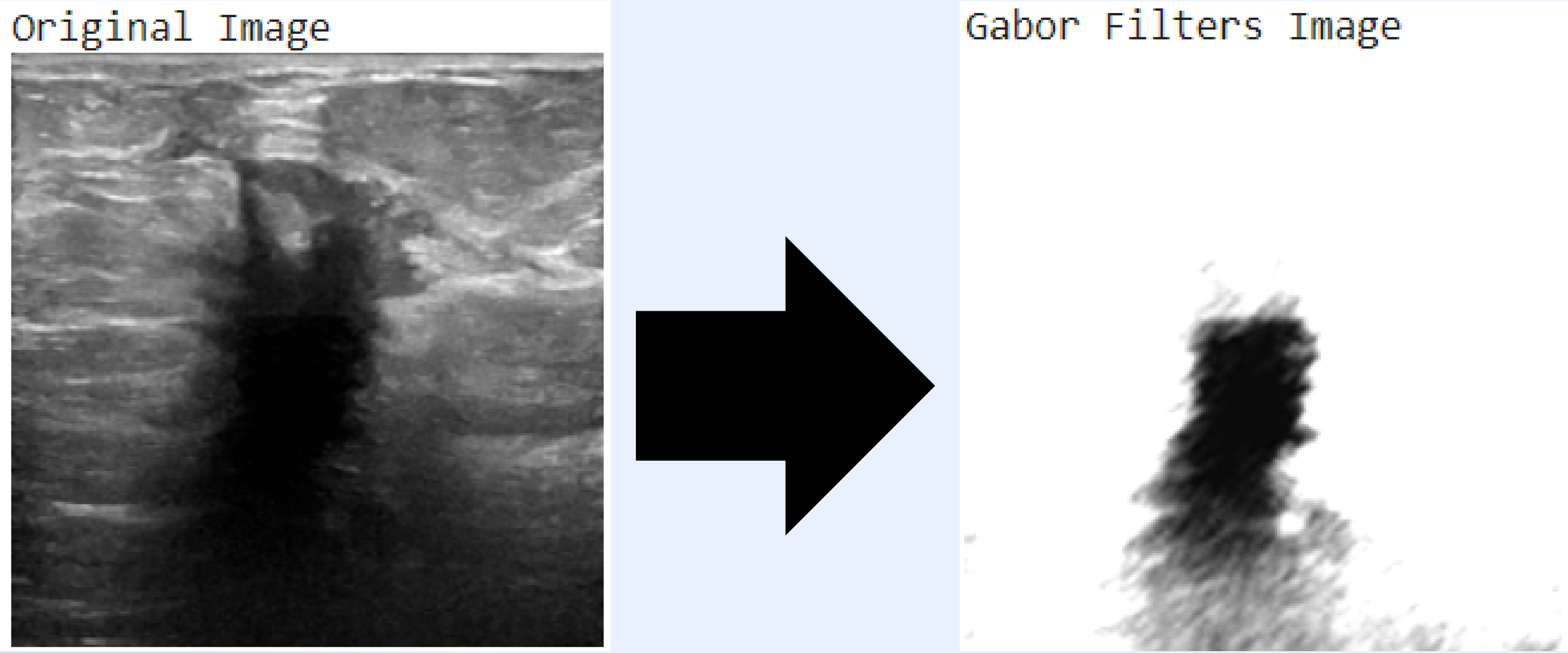
Gabor Filter







Gabor Filter



Feature Extraction

Feature Extraction

- **Feature extraction** is a critical step in image processing to transform raw image data into meaningful features to be used as inputs for machine-learning models.
- Our **feature extraction process** starts by iterating over its flattened pixel values for each image and saving them as separate features in the training and testing datasets. These features are used as inputs for the machine learning model. (***Pixel Intensity-Based Features***)
- The process was done using **os** and **cv2** libraries.

Model's Experiments

Original Pixel Values

Experiments / ML Classifiers	**100X100** PCA 100	**100X100** PCA 100 Augmenter
Random Forest	0.55	0.85
Multinomial NB	0.45	0.59
Support Vector Classifier	0.54	0.82
K-Nearest Neighbors	0.60	0.77
XG Boost	0.67	0.80
Light GBM	0.69	0.81

Original Pixel Values

Experiments / ML Classifiers	**100X100**	**200X200**	**150X150** Cropping 200X200	**100X100** PCA 80	**200X200** PCA 80	**200X200** Reflection
Random Forest	0.63 **TEST: 0.7**	0.61	0.63	0.66	0.66	0.75
Gradient Boosting	0.60	0.62	0.66	0.68	0.68	0.78
Histogram-based Gradient Boosting	0.65 **TEST: 0.7**	0.65 **TEST: 0.73**	0.64	0.69	0.75	0.79
K-Nearest Neighbors	0.63	0.63	0.61	0.57	0.71	0.64

Image Filtering

Experiments / ML Classifiers	**200X200** Gabor Filter	**200X200** Reflection & Gabor Filter	**200X200** Bilateral Filter	**200X200** Scharr Filter
Random Forest	0.57	0.63	0.65	0.52 **TEST: 0.66**
Gradient Boosting	0.59	0.68	0.64	0.50
Histogram-based Gradient Boosting	0.57	0.64	0.58 **TEST: 0.68**	0.58
K-Nearest Neighbors	0.65	0.54	0.60	0.60

Hyperparameters Tuning

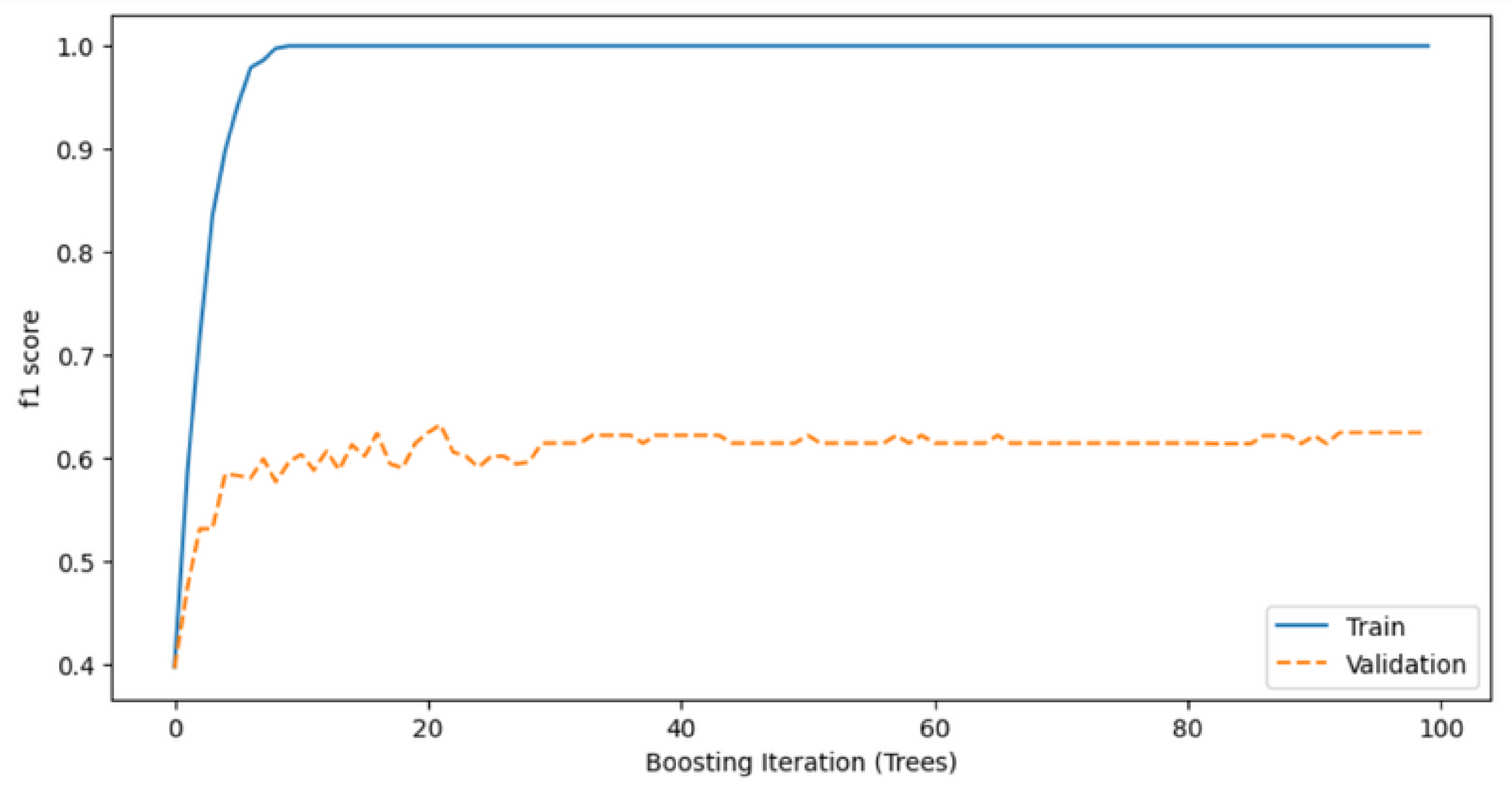
GridSearch CV:

- Random Forest Classifier: f1Score = 0.7
- Histogram-Based GBClassifier: f1Score = 0.65



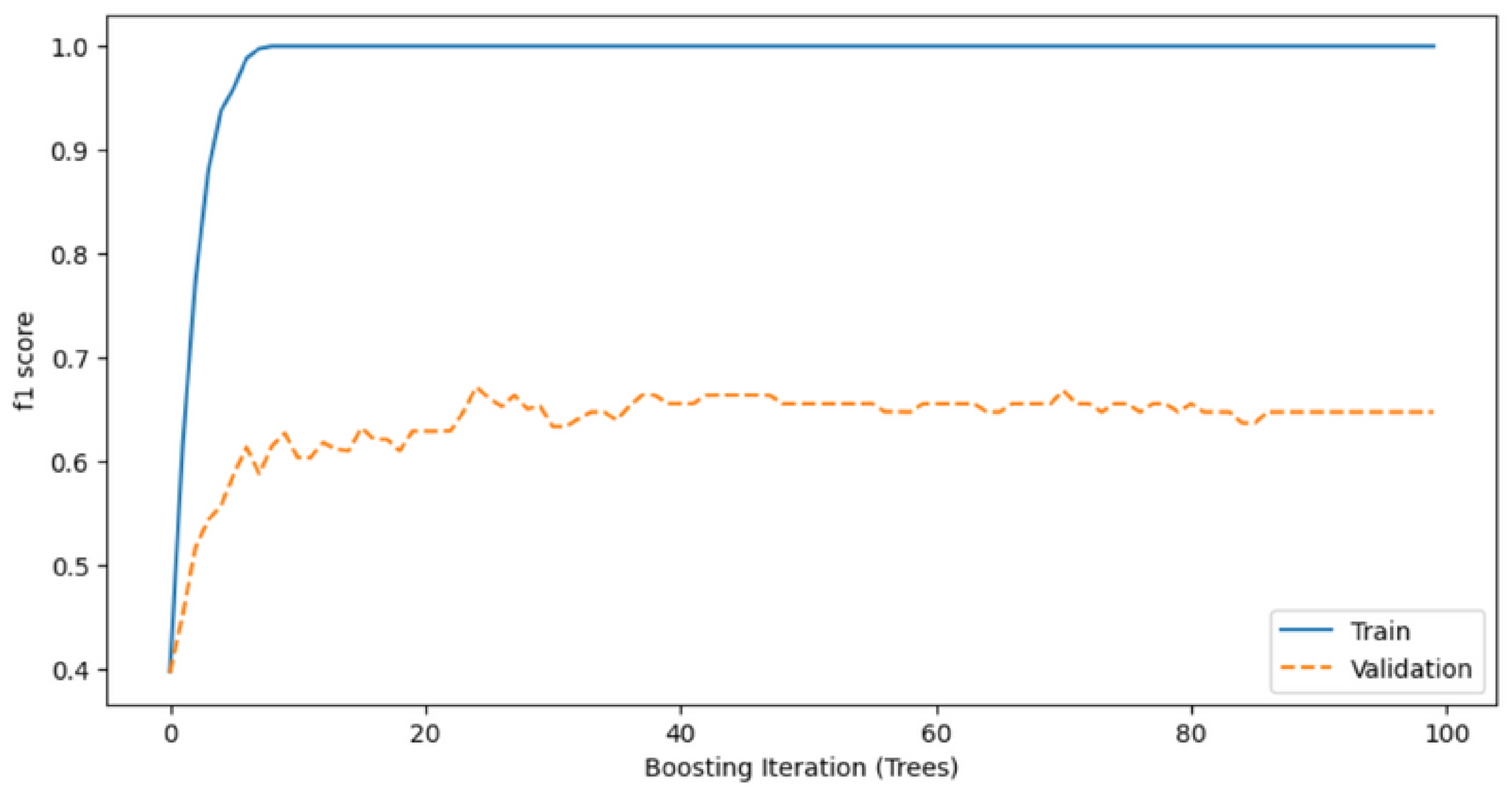
Learning Curves

Histogram-based Gradient Boosting Classifier



100X100

Histogram-based Gradient Boosting Classifier



200X200

Future Work

- Deep Learning Based Methods: CNN, which is the most accurate and extensively used model for breast cancer detection.
- Segmentation: Iterating through all the slices of an image and drawing a contour at the boundary.
- The number of pixel values in an image (also known as image resolution or size) can impact how images are preprocessed.

Conclusions

Histogram-based Gradient Boosting was the best classifier over all the used classifiers; it gave us a 0.73 f1-score by resizing the images as 200X200.

In conclusion, the best ML classifiers for the Breast Tumor Detection Problem are Boosting classifiers.

References

- Ensemble Flower-Pollination Algorithm for Automatic Breast Cancer Classification using Haar Features. S R Sannasi Chakravarth. et al.

- Explainable Ensemble Machine Learning for Breast Cancer Diagnosis Based on Ultrasound Image Texture Features. Alireza Rezazadeh et al. 2022.

THANK YOU